

Search Agents Assisted by Wireless Sensor Networks: Cooperation Algorithms and Performance

Nada Abughanam^{1,2}, Tamer Khattab¹ and Amr Mohamed²

¹IPL Lab, Electrical Engineering, Qatar University, Doha, Qatar

²Computer Science and Engineering, Qatar University, Doha, Qatar

Emails: nada.abughanam@qu.edu.qa, tkhattab@ieee.org, amrm@qu.edu.qa

Abstract—For any emergency, such as a toxic gas leakage, a timely response is essential to minimizing the damage it could cause. In this scenario, rather than endangering trained animal or human searchers, a localizing robot searcher can be deployed to find the original source of leakage. The deployed searcher can be a single mobile searcher, multiple cooperating mobile searchers, or a stationary Wireless Sensor Network (WSN) that localizes the source via distributed cooperation strategies. The use of multiple searchers can significantly improve the accuracy and efficiency of the search, but it is also more costly. In this paper, we study the effect of assisting a single mobile searcher with a WSN that is distributed around the environment. Utilizing the Infotaxis searching strategy, two algorithms are proposed to facilitate the cooperation between the searcher and the WSN nodes. The results show that even with only 16 WSN nodes in a 50x50 meter area, both models improve the success rate of the single searcher by over 65%, and are comparable to the performance of two searchers.

Index Terms—olfactory search, odor source localization, infotaxis, wireless sensor network.

I. INTRODUCTION

In emergency situations such as a toxic gas leak or a natural disaster, it is imperative to identify the location of the source of the gas and the location of the victims in order to prevent fatal situations. Olfactory search is a process that is widely observed in animals where they can find the source of stimulus (i.e. food, mate, or predator) released in the environment using their sense of smell. This process has been adapted for robot searchers to perform. The use of robotic searchers to localize the source of phenomena has been increasing due to the advances in gas detection and signal processing technology over the years. In addition, robotic searchers can be deployed in various turbulent or hazardous environments at low cost without worry regarding their safety and fatigue. Thus, some dangerous conditions (i.e. natural disasters, underground leaks, underwater leaks, etc) can make it such that a robotic searcher is the most logical choice. Inspired by the olfactory search of microorganisms and animals, odor source localization is used to find chemical sources where the substance emitted by the source is affected by the speed and direction of wind,

resulting in a plume of that substance in the environment. The process is divided loosely into plume finding, plume traversal, and source declaration. Plume finding involves the searcher attempting to locate the edges of the plume, where the searcher then traverses the plume by moving upwind across the plume. When the searcher is close to the source, the searcher declares that the source is found. Over the years, a number of odor search algorithms have been developed including gradient-based algorithms, bio-inspired algorithms, heuristic algorithms, inference-based algorithms, and learning algorithms. Gradient-based algorithms are inspired by smaller organisms, such as the *E. coli* bacteria, that follow the concentration gradient of the odor [1], [2]. The gradient is found by the robotic searcher either by measurements of different sensors placed on it or by measurements of different positions of the searcher. Meanwhile, bio-inspired algorithms draw inspiration from the behavior of larger organisms observed when they are searching for food or predators, including moths and lobsters [3], [4]. Heuristic algorithms [5] include formation-based methods and Particle Swarm Optimization (PSO) methods. Based on multiple searchers, these algorithms consider Odor Source Localization (OSL) as an optimization problem. Further, OSL algorithms based on probabilistic inference have been proposed. These methods utilize odor and gas dispersion models to localize the source of that odor [6], [7]. By collecting sparse measurements, a probability distribution of that odor source is created. One such algorithm is Infotaxis, proposed by Vergassola et al. [8], which utilizes information-driven principles. Infotaxis uses concentration measurements to model a probability distribution of the location of the source, where the robot searcher moves to locations with more information about the source to reduce the entropy of the distribution. Finally, learning algorithms utilize deep learning (DL) and reinforcement learning (RL) to find the odor source location [9], [10].

To improve the performance, many of these single searcher algorithms were adapted for multiple searchers [11]. The use of multiple searchers can increase the odds of finding the source location in addition to decreasing the time it takes to locate the source. However, it can be more cost-effective to use a single searcher that is assisted by a Wireless Sensor

This research was made possible by grant number QUCG-CENG-22/23-600 from Qatar University. The research work of T. Khattab was supported by QRDI grant number AICCO3-0530-200033 from QNRF. The statements made herein are the sole responsibility of the authors.

Network (WSN) to localize the source with a success rate that is comparable to two searchers. Previous work addressed the use of a WSN to localize the source of odor, where the WSN nodes collaborate. However, the collaboration between a moving searcher with its own sensing capabilities and a WSN was not addressed.

In this paper, a single searcher assisted by a WSN to localize the odor source is investigated. The searcher employs infotaxis as the search movement strategy, while utilizing different cooperation algorithms between the WSN and a single search agent to improve the agent's performance.

The remainder of this paper is organized as follows: Section II summarizes the related work relevant to the topic. Section III illustrates the gas dispersion model, the source searching strategy, and the communication model. Two cooperation algorithms are proposed in Section IV. Section V discusses the parameters used in the simulation and compares the results of several simulations. Finally, Section VI concludes the work and describes future work.

II. RELATED WORK

Previous research mostly focused on proposing different algorithms and strategies for using WSNs in localizing the source of some substance. However, WSNs have not been used in the context of assisting a moving searcher by providing static sensing data from various locations around the environment. Additionally, although collaboration between a WSN and a moving searcher has been discussed in contexts other than gas source localization, these works consider that the mobile searcher lacks sensing capabilities and exclusively relies on the sensing of the WSN as in [12]. The following describes different odor source localization methods for WSNs.

Zhao and Nehorai [13] proposed a distributed sequential Bayesian estimation method that uses WSNs to localize the source of some substance, find the distribution of concentration of the source, and predict the evolution of its plume. The distribution of substance concentration is modeled by the diffusion equation. Then, to find the source location, the state belief is transmitted to each node sequentially and updated with the new sensor information sampled by each node. Two belief representation methods were implemented, Gaussian density approximation and linear combination of polynomial Gaussian density functions (LPG function) approximation.

Li et al. [14] put forward a source localization algorithm that combines particle filter and the weighted centroid method to find the source of gas. Due to the nonlinearity of gas diffusion models in turbulent wind, the traditional weighted centroid method in a WSN inaccurately estimates the location of the odor source. In the proposed algorithm, the weighted centroid method is used to estimate the source location which is then used as input to the particle filter. This improves the speed of convergence of the particle filter and reduces the localization error that would be produced by the weighted centroid method.

Cao et al. [15] proposed a distributed and efficient gas source localization algorithm using WSNs, the Distributed Sequential Convex Combination (DSCC) method. In DSCC,

each node in the network is successively selected to perform the source location estimation. Once a node is selected, the overall estimate from the previous node is received and the node performs its own individual estimation, which is combined with the overall estimate via convex combination. The process is then repeated by the node selecting the next node and sending its updated overall estimate to the next node.

Zhang et al. [16] proposed a gas source localization algorithm that reduces the data transmission and resource consumption that occurs in wireless sensor networks when used to perform gas source localization. This consumption of resource is due to requiring a multitude of redundant sampling information from various locations in the environment to locate a single gas source. Using compressed sensing theory, the environment was divided into a large-scale global environment and a small-scale local environment. Initially, following the global grid division, variational Bayesian inference is used to jointly estimate the source distribution and the initial source location in a candidate grid. Then, within the local grid division, different sensor node clusters utilize parallel variational Bayesian inference to estimate the position of the source and environment parameters.

III. SYSTEM MODEL

A single agent using the infotaxis strategy is deployed in a finite grid area with a single odor source. To improve the performance of the searcher, a WSN which is already deployed in the grid is leveraged by the mobile searcher as shown in Fig. 1. The distribution of the nodes is uniform and equidistant, where the distance between each node is equal. The WSN nodes send their sensing information to the searcher via a communication channel that suffers from errors and path-loss. The searcher then utilizes the information received from the nodes using the multi-agent infotaxis strategy, where it uses the information from the WSN nodes the same way it would for multiple searchers. The source is assumed to be easily identifiable by the searcher, where the source is declared as found if the searcher is within a short distance from the source. Additionally, the source is declared as not found if a certain number of steps (or time period) is passed without localizing the source. In this section, the source dispersion model, searching strategy, and the communication model are described in more detail.

A. Dispersion Model

For an odor source placed in a grid, the odor spreads around the environment into different concentrations following a propagation model. The average concentration $C(\mathbf{r}|\mathbf{r}_0)$ of the odor at location $\mathbf{r}$ given that the source is positioned in $\mathbf{r}_0$ must satisfy the Advection-Diffusion equation [8]. This equation is used to simulate how a plume that consists of odor particles spreads from the source in a turbulent environment. Specifically, how advection and diffusion affect the movement of odor particles when placed in a medium. The source releases particles at a rate R with an average lifetime of τ and the particles diffuse with diffusivity D , affected by wind

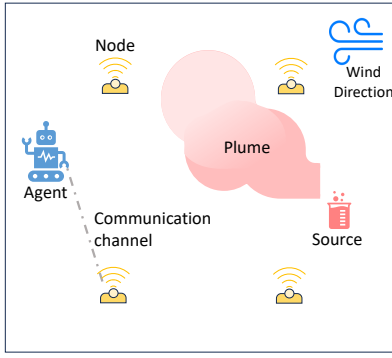


Fig. 1: System model for a WSN-assisted searcher.

that is blowing in the y -direction with an average wind speed V . The solution for the Advection-Diffusion equation allows the calculation of the expected hit rate $R(\mathbf{r}|\mathbf{r}_0)$ given by Eq. 1, where K_0 is the Bessel function of zeroth order and a is the searcher's radius of search. The direction of the wind affects the linear distance between the location of the searcher and the location of the source, $-(y-y_0)$. For instance, for upwind, the term uses the negative of the distance on the x -axis, whereas downwind uses the distance on the x -axis.

$$R(\mathbf{r}|\mathbf{r}_0) = \frac{R}{\ln(\frac{\lambda}{a})} e^{\frac{-(y-y_0)V}{2D}} K_0\left(\frac{|\mathbf{r}-\mathbf{r}_0|}{\lambda}\right), \quad \lambda = \sqrt{\frac{D\tau}{1 + \frac{V^2\tau}{4D}}}. \quad (1)$$

B. Source Search Strategy

The searcher uses the Infotaxis searching strategy proposed by [8]. The searcher will first build a priori probability distribution with uniform probability, where the probability of finding the source at any location in the search area is equal. Using measurements of the odor concentrations in the environment, the probability distribution is updated until the source is found or a specified amount of time passes. These measurements are used to collect detection events, wherein when the searcher is able to detect a number of particles, it is counted as a 'hit' event. Otherwise, it is counted as a 'miss' event. For the sake of simplicity, any number of hit events are counted as one hit, and miss events are counted as zero hits. This process is modeled by a Poisson distribution with the mean being the expected mean hit rate.

The searcher combines its own measurements with that of several stationary WSN nodes in the same way as it would for multiple agents in [17]. The posterior probability distribution is obtained as shown in Eq. 2.

$$P_{t+dt}(\mathbf{r}_0) = \frac{P_t(\mathbf{r}_0) \prod_{i=1}^n e^{-R(\mathbf{r}_i(t+dt)|\mathbf{r}_0)dt} R^{k_i}(\mathbf{r}_i(t+dt)|\mathbf{r}_0)}{Z_{t+dt}}, \quad (2)$$

This follows the Bayesian framework, with the prior probability distribution being denoted as $P_t(\mathbf{r}_0)$ at time t for a source that is located at $\mathbf{r}_0$. The posterior distribution is updated using the information of a hit or miss detected by the searcher and the WSN nodes i to n during time step dt . Z_{t+1} is the normalizing constant.

In this paper, the control decision for the searcher is limited to up, down, left, right, and stay. The searcher makes the decision to move to the location that results in the highest reduction of uncertainty regarding the location of the source, found via Shannon's entropy. Equation 3 is used to find the expected change in entropy as a result of encountering a hit or miss if the searcher were to move to a potential location $\mathbf{r}_j$.

$$\Delta S(\mathbf{r} \rightarrow \mathbf{r}_j) = P_t(\mathbf{r}_j)(-S) + (1 - P_t(\mathbf{r}_j)) \left(\sum_{k=0}^1 \rho_k(\mathbf{r}_j) \Delta S_k \right), \quad (3)$$

where S is Shannon's entropy of the probability distribution, $P(\mathbf{r}_j)$ is the probability of finding the source in location $\mathbf{r}_j$, and $\rho_k(\mathbf{r}_j)$ is the probability of not finding the source while encountering k hits which is modeled as a Poisson distribution, as the hits are considered independent. ΔS_k is the change in entropy if the searcher does not find the source but detects $k = [0, 1]$ hits. The searcher will repeat the process until the source is found or some amount of time passes without finding the source.

C. Communication Model

The message sent from the WSN to the searcher only contains the status of a hit or miss (1 or 0). Since the WSN nodes are stationary, the searcher knows the location of the nodes throughout the search and is equipped with a localization device to stay aware of its own location. The communication channel between the searcher and WSN nodes is modeled by a log-distance path loss model. Equation 4 models the effect of path loss and fading.

$$PL(d) = PL(d_0) + 10\gamma \log\left(\frac{d}{d_0}\right) + X_g, \quad (4)$$

Where $PL(d)$ is the total path loss for a total path length d , $PL(d_0)$ is the path loss at reference distance d_0 , γ is the path loss exponent, and X_g is a normal random variable that represents shadowing with a variance σ_{X_g} . The path loss model will give an indicator as to the power of the signal at the receiving agent, if the signal suffers from a large amount of attenuation, the receiver will not receive the signal. Additionally, each message sent through the channel will have a certain probability of error P_e . If the message suffers from error, the shared hit/miss information will be flipped (from 1 to 0 or 0 to 1).

IV. PROPOSED COOPERATION ALGORITHMS

Under the assumption that the searcher uses the same data fusion model used by the multi-agent strategy, for the searcher to gain the most benefit from the deployed WSN, it must be able to choose which nodes it gathers measurement information from. It is not useful to collect data from all the nodes in the WSN due to the searcher being overwhelmed and distracted with a large amount of unnecessary information. Since the WSN nodes have fixed locations, the searcher would be receiving redundant information about the changes

in the environment in the same locations, instead of the nodes guiding it through the search space. As such, some selection methods must be implemented for the searcher. Two approaches are proposed for the selection criteria for the searcher, detailed below.

A. Closest Only

This approach involves the searcher only receiving information from nodes that are located within a certain radius around the searcher. This somewhat emulates a scenario where the searcher is equipped with extended arms that can sample the environment from different locations around it. This is more clearly shown in Fig. 2.

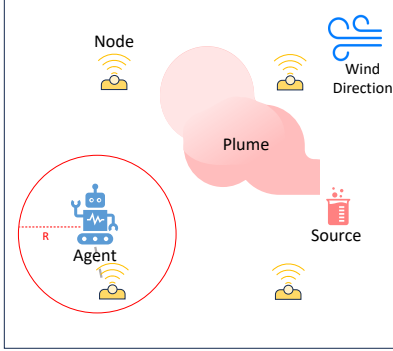


Fig. 2: Illustration of the closest only data sharing algorithm.

Given that the searcher knows its own location and the locations of all the WSN nodes which are placed in fixed known locations, the searcher communicates with all nodes that are located within a certain radius around it when it stops to make its own measurement. The searcher then combines its own measurement with the measurements received from the nodes using the multi-agent infotaxis strategy to update its probability distribution of the source location. The subsequent movement decision is made as normal. The searcher can gather more measurement information by either increasing the number of nodes, increasing the radius around the searcher, or increasing both.

B. Pseudo-moving WSN

The second approach only allows the searcher to receive data from a single node. The selection method for this node entails the searcher performing calculations for the node as it would if the node were a moving searcher. By calculating the direction that a searcher in that location would move in order to decrease the entropy, the searcher selects the next node that is located in that direction, as shown in Fig. 3. If no nodes are present in that direction, the searcher selects a random node instead.

This approach attempts to mimic a second searcher that hops on sensor locations as it moves. After randomly selecting the initial node and receiving its measurement information, the decrease in entropy around the node is calculated by the searcher, where it decides the direction that provides the largest

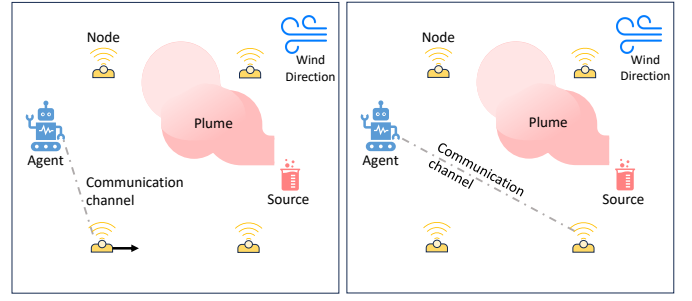


Fig. 3: Illustration of the pseudo-moving data sharing algorithm.

TABLE I: Simulation parameters.

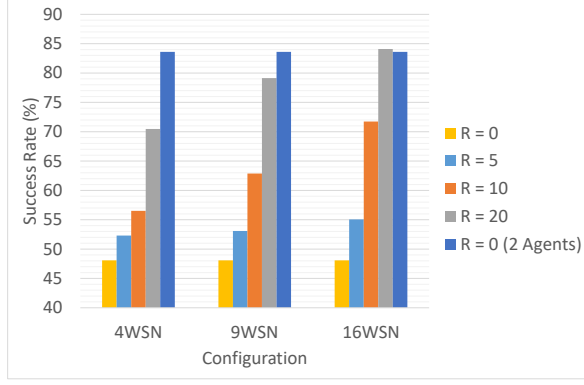
Notation	Description	Value
V	Average wind speed	1 m/s
D	Diffusion coefficient	1 m ² /s
R	Release rate	5 Hz
τ	Average life of particles	2500 s
a	Radius of searcher	1 m
Radius	Source declaration radius	1 m
γ	Path-loss exponent	1.6
σ_{X_g}	Shadowing variance	4.5
P_e	Probability of error	0.02
P_{TX}	Transmit power	8 dBm
P_{RX}	Receiver sensitivity	-98 dBm

decrease in entropy similar to what a moving searcher would do. The searcher then begins communicating with the node located in that direction.

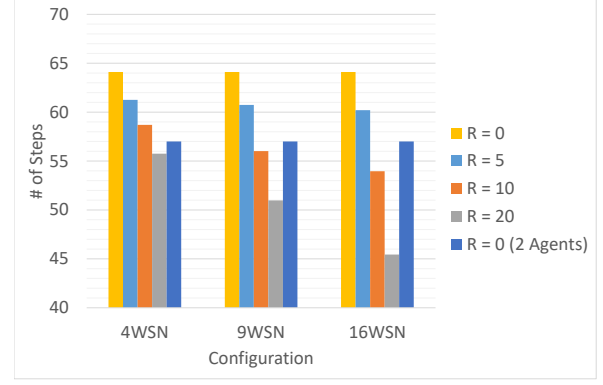
As the number of nodes increases in this approach, the searcher is able to communicate with a larger variety of nodes, decreasing the occurrence of ping-ponging where the searcher is stuck communicating between two nodes back and forth.

V. PERFORMANCE EVALUATION

Each scenario was simulated and averaged over 5,000 runs, where the location of the source and the searcher in addition to the wind direction were randomized. The searcher moves at a speed of 1 meter/second. For both scenarios, the WSN nodes are distributed in fixed locations that are equidistant to each other. The source is easily identifiable and is considered found if the searcher is within 1 meter of the source. The simulation ends if the searcher finds the source, or if the number of steps taken by the searcher exceeds a certain threshold. This threshold is 100 steps for the 50x50 meter grid, 200 steps for the 100x100 meter grid, etc. The searcher communicates with the WSN nodes via an ad hoc communication protocol e.g. ZigBee due to its low power consumption and effective range [18]. The messages from the nodes are transmitted with a transmission power P_{TX} of 8dBm where the agent has a receiver sensitivity P_{RX} of -98dBm. The simulation parameters are summarized in Table I, where the probability of error is fixed at 0.02. The error affects the message by flipping the hit information, and the path-loss can cause the message to be dropped.

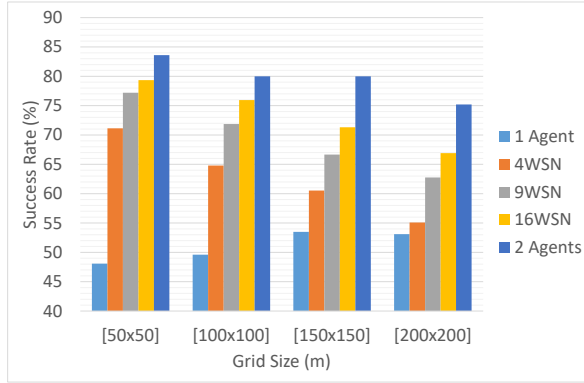


(a) Success rate.

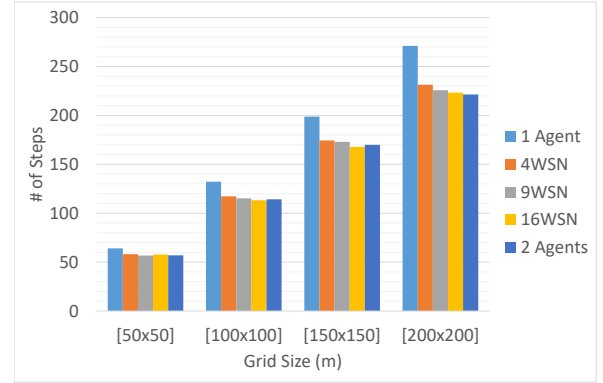


(b) Number of steps.

Fig. 4: The success rate and number of steps of the searcher in the first scenario when supported by 4, 9, and 16 WSN nodes where communication is allowed within a radius of 0 (for one and two agents), 5, 10, and 20 meters in a 50x50 meter grid.



(a) Success rate.



(b) Number of steps.

Fig. 5: The success rate and number of steps of the searcher in the second scenario when supported by 4, 9, and 16 WSN nodes in different grid sizes.

To evaluate the performance of the searcher, the success rate and the average number of steps taken by the searcher in each scenario are compared. The performance of the single searcher without the assistance of the WSN is used as a benchmark. The performance is evaluated over multiple grid sizes, varying from 50x50 meters to 200x200 meters. The number of WSN nodes is varied from 4 nodes to 16 nodes.

For the first scenario where the searcher is assisted by the nodes that are within a specified radius, the results are shown for a grid size of 50x50 meters. The radius R is varied between 5 meters to 20 meters. Fig. 4 (a) shows the success rate, and Fig. 4 (b) shows the number of steps taken by the searcher.

It can be seen that as the number of nodes increases, the success rate of the searcher similarly increases, while the number of steps taken by the searcher decreases. The

same improvement in performance is observed when the communication radius around the searcher is increased, where the searcher is assisted by a larger number of nodes. When the radius is 0 meters, it is equivalent to the searcher not being assisted by any nodes and performing the search alone. At the maximum number of nodes and radius, the success rate of the searcher is increased by 75% and the number of steps is decreased by 29%.

Meanwhile, in the second scenario, the success rate of the searcher assisted by 4 to 16 WSN nodes in various grid sizes is shown in Fig. 5 (a). The number of steps associated is displayed in Fig. 5 (b). As the number of nodes increases, the WSN is able to emulate a moving searcher more successfully with a lower step size. In the case of the 50x50 meter grid, the success rate is increased by 65% as compared to a single agent,

TABLE II: The success rate of different configurations in different grid sizes.

Grid (m)	1 Agent	2 Agents	Closest (16 WSN)	Pseudo-moving (16 WSN)
[50x50]	48.08%	83.60%	84.10%	79.35%
[100x100]	49.60%	80%	68.2%	75.95%

and is only lower than the case of two moving searchers by 5%. As the number of nodes increases further, the performance of the single searcher assisted by a pseudo-moving WSN can exceed the performance of two searchers.

Table II shows the difference in success rate between a single searcher, two searchers, a single searcher assisted by the closest of 16 nodes within a 20 meter radius, and a single searcher assisted by a pseudo-moving 16 node WSN in a 50x50 meter grid and a 100x100 meter grid. Although the closest WSN is superior to the pseudo-moving WSN in the 50x50 grid, its performance degrades as the grid size increases. This is due to the closest WSN requiring the searcher to be close to a node to be effective, while the pseudo-moving WSN can move between the nodes to assist the searcher giving it more flexibility.

VI. CONCLUSION

In this paper, an OSL searcher that is assisted by a WSN was implemented. Assisting the searcher with a WSN greatly improves the performance of the searcher and brings it up to par with two agents. This applies to both proposed cooperation algorithms. The closest only algorithm has a higher performance than the pseudo-moving algorithm for a smaller grid size, but it is more affected by the increase in the size of the grid for the same number of nodes. Meanwhile, the pseudo-moving algorithm is more resilient and maintains its success rate even when the grid size is increased. The choice between the two algorithms depends on the number of WSN nodes deployed given the size of the search area. Future work will aim to improve upon the proposed algorithms in addition to implementing them in the real world to observe their actual feasibility.

REFERENCES

- [1] R. A. Russell, D. Thiel, R. Devezza, and A. Mackay-Sim, "A robotic system to locate hazardous chemical leaks," in *Proceedings of 1995 IEEE International Conference on Robotics and Automation*, vol. 1. IEEE, 1995, pp. 556–561.
- [2] G. Ferri, E. Caselli, V. Mattoli, A. Mondini, B. Mazzolai, and P. Dario, "Spiral: A novel biologically-inspired algorithm for gas/odor source localization in an indoor environment with no strong airflow," *Robotics and autonomous systems*, vol. 57, no. 4, pp. 393–402, 2009.
- [3] P. B. Reeder and B. W. Ache, "Chemotaxis in the florida spiny lobster, *panulirus argus*," *Animal Behaviour*, vol. 28, no. 3, pp. 831–839, 1980.
- [4] H. Ishida, K. Hayashi, M. Takakusaki, T. Nakamoto, T. Moriizumi, and R. Kanzaki, "Odour-source localization system mimicking behaviour of silkworm moth," *Sensors and Actuators A: Physical*, vol. 51, no. 2-3, pp. 225–230, 1995.
- [5] J. M. Soares, A. P. Aguiar, A. M. Pascoal, and A. Martinoli, "A distributed formation-based odor source localization algorithm-design, implementation, and wind tunnel evaluation," in *2015 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2015, pp. 1830–1836.
- [6] J. A. Farrell, S. Pang, and W. Li, "Plume mapping via hidden markov methods," *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, vol. 33, no. 6, pp. 850–863, 2003.
- [7] S. Pang and J. A. Farrell, "Chemical plume source localization," *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, vol. 36, no. 5, pp. 1068–1080, 2006.
- [8] M. Vergassola, E. Villermaux, and B. I. Shraiman, "'infotaxis' as a strategy for searching without gradients," *Nature*, vol. 445, no. 7126, pp. 406–409, 2007.
- [9] H. Kim, M. Park, C. W. Kim, and D. Shin, "Source localization for hazardous material release in an outdoor chemical plant via a combination of lstm-rnn and cfd simulation," *Computers & Chemical Engineering*, vol. 125, pp. 476–489, 2019.
- [10] H. Hu, S. Song, and C. P. Chen, "Plume tracing via model-free reinforcement learning method," *IEEE transactions on neural networks and learning systems*, vol. 30, no. 8, pp. 2515–2527, 2019.
- [11] A. Francis, S. Li, C. Griffiths, and J. Sienz, "Gas source localization and mapping with mobile robots: A review," *Journal of Field Robotics*, vol. 39, no. 8, pp. 1341–1373, 2022.
- [12] S. Beyme and C. Leung, "Wireless sensor network-assisted, autonomous mapping with information-theoretic utility," in *2014 IEEE 6th International Symposium on Wireless Vehicular Communications (WiVeC 2014)*. IEEE, 2014, pp. 1–5.
- [13] T. Zhao and A. Nehorai, "Distributed sequential bayesian estimation of a diffusive source in wireless sensor networks," *IEEE Transactions on Signal Processing*, vol. 55, no. 4, pp. 1511–1524, 2007.
- [14] Q. Li, Z. Liu, and X. Xiao, "A gas source localization algorithm based on particle filter in wireless sensor network," *International Journal of Distributed Sensor Networks*, vol. 11, no. 11, p. 874532, 2015.
- [15] M.-L. Cao, Q.-H. Meng, Y.-Q. Jing, J.-Y. Wang, and M. Zeng, "Distributed sequential location estimation of a gas source via convex combination in wsns," *IEEE Transactions on Instrumentation and Measurement*, vol. 65, no. 6, pp. 1484–1494, 2016.
- [16] Y. Zhang, T. Wang, Y. Shi, and L. Zhang, "Joint variational bayesian based localization estimation algorithm on distributed gas source sensor network," *Computer Communications*, vol. 154, pp. 262–268, 2020.
- [17] J. Masson, M. B. Bechet, and M. Vergassola, "Chasing information to search in random environments," *Journal of Physics A: Mathematical and Theoretical*, vol. 42, no. 43, p. 434009, 2009.
- [18] K. F. Haque, A. Abdelgawad, and K. Yelamarthi, "Comprehensive performance analysis of zigbee communication: an experimental approach with xbee s2c module," *Sensors*, vol. 22, no. 9, p. 3245, 2022.